PEC
EXPLORE INNOVATE EXCEL

Zigbee based Machine to Machine Communication for On-road Low-visibility Collision Prevention

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Under the Guidance of
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Major Project I – End Semester Evaluation

Introduction

This project utilizes Zigbee-enabled ESP32-H2 microcontrollers for vehicle-to-vehicle communication, ensuring reliable data exchange of GPS and motion data in low-visibility conditions, enhancing safety, situational awareness, and collision avoidance efficiently.

Objectives

1. Vehicle-to-vehicle (V2V) communication to enhance **transportation safety and efficiency**, especially in adverse conditions like snowstorms or smog where traditional navigation aids often fail.
2. Zigbee, a low-power and cost-effective protocol, to be integrated with ESP32-H2 micro controllers to **establish reliable short-range wireless communication between vehicles.**
3. To use GPS modules (Neo-6M) for **real-time location tracking and acceleration monitoring** (MPU-6050) for motion monitoring, ensuring precise and timely data exchange.

Real Life scenarios for Zigbee V2V Communication

01 Foggy Highway Collisions

02 Snowstorm Blind Spot Alerts

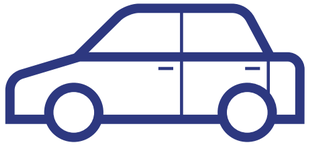
03 Dust Storm Driving

04 Rain-Induced Skidding

Why Not LiDAR, CAMERA, or Wi-Fi?

Technology	Limitations	Why Zigbee Excels
LiDAR	**High cost**, making it unsuitable for widespread adoption. Performance degrades in fog, rain, or snow due to **laser scattering.**	**Low cost and scalable** for mass deployment. Reliable in adverse weather conditions.
CAMERA	They can be **blinded by bright lights** and see reflections as real images. In addition, **images are obscured** by rain, fog and snow.	**Enhances ADAS reliability** by ensuring that critical signals (e.g., proximity alerts) are transmitted despite visual obstructions.
Wi-Fi	**High power consumption unsuitable** for vehicles relying on low-energy systems. **Requires higher bandwidth**, making it costlier.	Zigbee's **low power consumption** ensures energy efficiency. Optimised **for small data packets** and local networks.

Necessary Data for Communication



Motion Data (From MPU-6050):

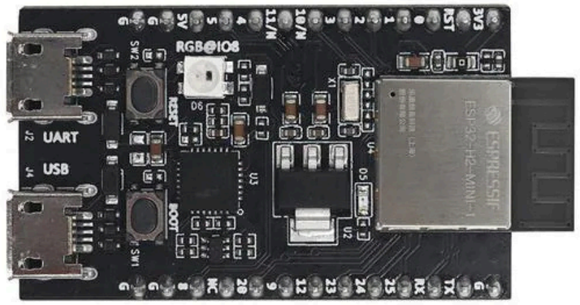
- **Acceleration**: Detects sudden changes in speed, such as hard braking.
- **Angular Velocity**: Tracks rotational movements like sharp turns or skidding.
- Helps **identify erratic vehicle behavior** in low-visibility conditions.

Positional Data (From NEO-6M GPS):

- **Location**: Provides latitude and longitude for real-time vehicle tracking.
- **Direction**: Monitors movement to identify the direction of moving vehicles.
- Enables proximity **awareness and situational clarity** in dense traffic.

Sensors & Microprocessors

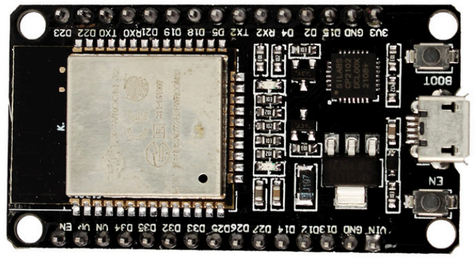
ESP32 H2



- **Connectivity**: The ESP32-H2 supports Zigbee 3.0 enabling low-power mesh networking for IoT application
- **Use case:** To transmit data wirelessly using Zigbee protocol from one node to another.

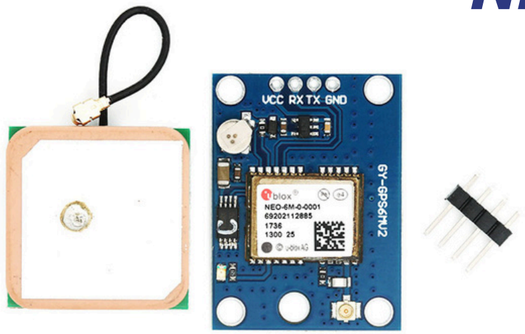
NODE MCU ESP32

- **I2C and UART Protocols:** The ESP32 supports I2C and UART for communication between devices within an embedded system.
- **Use Case:** To process and acquire data from sensors.



Sensors & Microprocessors

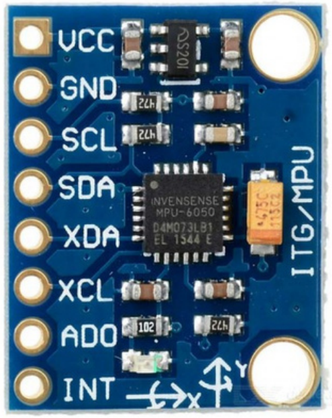
NEO-6M GPS



- **GPS**: The NEO-6M is a compact and reliable receiver designed to provide accurate real-time location and time information.
- **NMEA**: The NEO-6M provides positional data in NMEA (National Marine Electronics Association) format.

MPU 6050 Accelerometer

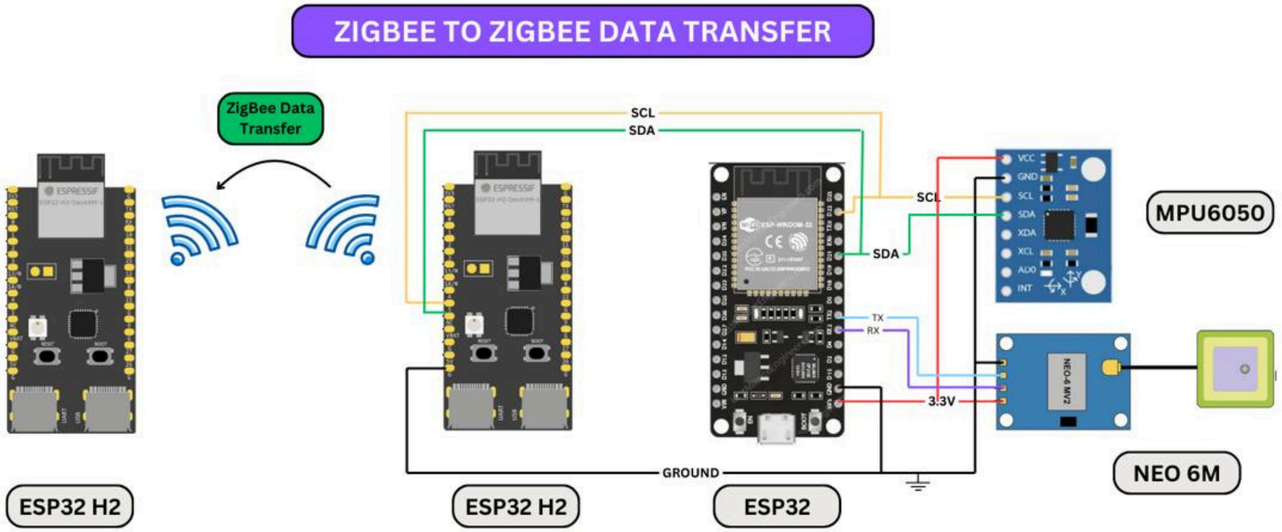
- **Acceleration and Gyroscope**: The MPU-6050 is a motion-tracking sensor that integrates a 3-axis accelerometer and a 3-axis gyroscope.
- **DMP**: The MPU-6050 uses a Digital Motion Processor (DMP) to combine accelerometer and gyroscope data.



System Design

- **Each vehicle** is equipped with an ESP32 controller and a Zigbee module for V2V communication.
- Acceleration data is captured using the **MPU 6050 sensor (I2C), and GPS data using the NEO 6M module (UART).**
- **ESP32 processes sensor data** and exchanges critical information via Zigbee when vehicles are nearby.
- Shared data includes location and speed, enabling **real-time collision alerts.**

Connection Diagram



NEO-6M GPS Module	ESP32
VCC	3V3
RX	TX2 (GPIO 17)
TX	RX2 (GPIO 16)
GND	GND

MPU6050 Module	ESP32
VCC	3V3
SCL	GPIO 22
SDA	GPIO 21
GND	GND

ESP32 Master	ESP32 H2 Slave
SDA (GPIO 21)	SDA (GPIO 5)
SCL (GPIO 22)	SCL (GPIO 4)
GND	GND

On-site Testing and Validation

	Sender	Success Rate (%)	Receiver	Success Rate (%)
Without Obstacles	**500/500**	**100**	**498/500**	**99.60**
With Obstacles	**544/544**	**100**	**540/544**	**99.26**
On Road	**563/563**	**100**	**557/563**	**98.93**

Conducted in three environments:

- **Without Obstacles:** Controlled conditions to test short-distance performance.
- **With Obstacles:** Real-world scenarios with obstacles and moderate traffic.
- **On Road:** Long-distance, dynamic conditions to evaluate range and reliability.

Measured success rates, error rates, and consistency of data transmission.

Thank You

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